University of California, Los Angeles — Homework 1
Fall 2025

Math 151A: Applied Numerical Methods

1. **Integrability and Approximation**

   (a) Determine whether $f(x) = \sin(x^2)$ is Riemann integrable on $[0, 1]$. Justify your answer.
   Yes. The function $f(x) = \sin(x^2)$ is continuous on $[0, 1]$ thus it is uniformly continuous which implies Riemann integrable.

   (b) Let $f(x) = \sin(1/x^2)$ for $x \in (0, 1]$ and define $f(0) = 0$. Determine whether f is Riemann integrable on $[0, 1]$. Yes. Notice the function $f(x) = \sin(1/x^2)$ is not continuous at 0, but we can define the sequence which is tagged on the right-endpoint

   $$S_n = \frac{1}{n}\sum_{k=1}^{n} f\left(\frac{k}{n}\right).$$

   Fix $\varepsilon > 0$ and choose $\delta \in (0, 1)$ so that $\delta < \varepsilon/2$. Let $m = \lfloor n\delta \rfloor$. The contribution of the "near–0 block" $(0, \delta]$ to S_n is

   $$\frac{1}{n}\sum_{k=1}^{m} f\left(\frac{k}{n}\right),$$

   and since $|f| \le 1$,

   $$\left|\frac{1}{n}\sum_{k=1}^{m} f\left(\frac{k}{n}\right)\right| \le \frac{m}{n} \le \delta < \frac{\varepsilon}{2}.$$

   (c) Consider $f(x) = e^{-x^2}$ on $[0, 1]$. This function is continuous, hence integrable.
   Write a short algorithm that:

   - partitions $[0, 1]$ into n equal subintervals,
   - chooses a tag point t_i in each subinterval $[x_{i-1}, x_i]$,
   - computes the Riemann sum

     $$S_n = \sum_{i=1}^{n} f(t_i)\,(x_i - x_{i-1}),$$

   - and reports the value of S_n.

   Your algorithm should allow mesh refinement by increasing n. Run your code for $n = 4, 8, 16, 32$ and comment on how S_n behaves as the mesh is refined. Please attach your code and outputs to the homework submission.

   In Figure 1 we observe that the midpoint converges fastest, followed by random, then right then left.

2. **Taylor Polynomial Approximation**

   Find the third Taylor polynomial $P_3(x)$ for $f(x) = \sqrt{1 + x}$ about $x_0 = 0$. Use $P_3(x)$ to approximate $\sqrt{0.5}, \sqrt{0.75}, \sqrt{1.25}, \sqrt{1.5}$, and compute the absolute errors.

   $$\begin{aligned} f(x) &= (1 + x)^{1/2}, \\ f'(x) &= \tfrac{1}{2}(1 + x)^{-1/2}, \\ f''(x) &= -\tfrac{1}{4}(1 + x)^{-3/2}, \\ f^{(3)}(x) &= \tfrac{3}{8}(1 + x)^{-5/2}. \end{aligned}$$

```python
import math
import random

def f(x):
    return math.exp(-x*x)

def riemann_sum(n, tag="mid", seed=None):
    """
    Riemann sum of f on [0,1] using n equal subintervals.
    tag in {"left","right","mid","random"} chooses the tag point in each subinterval.
    """
    assert n > 0
    assert tag in {"left", "right", "mid", "random"}
    if seed is not None and tag == "random":
        random.seed(seed)

    a, b = 0.0, 1.0
    h = (b - a) / n
    s = 0.0
    for i in range(n):
        xL = a + i*h
        xR = xL + h
        if tag == "left":
            t = xL
        elif tag == "right":
            t = xR
        elif tag == "mid":
            t = 0.5 * (xL + xR)
        else:  # "random"
            t = random.uniform(xL, xR)
        s += f(t) * h
    return s

if __name__ == "__main__":
    Ns = [2**i for i in range(3,20)]
    I_exact = math.sqrt(math.pi)/2 * math.erf(1.0)  # ∫0^1 e^{-x^2} dx

    print("Riemann sums for f(x)=exp(-x^2) on [0,1]")
    print(f"Exact (via erf): {I_exact:.6f}\n")

    for n in Ns:
        S_left  = riemann_sum(n, tag="left")
        S_right = riemann_sum(n, tag="right")
        S_mid   = riemann_sum(n, tag="mid")
        S_rand  = riemann_sum(n, tag="random", seed=42)
        print(f"n={n:>2}:  left={S_left:.6f}  right={S_right:.6f}  mid={S_mid:.6f}  random={S_rand:.6f}")
```

```
Riemann sums for f(x)=exp(-x^2) on [0,1]
Exact (via erf): 0.746824

n= 8:  left=0.785373  right=0.706358  mid=0.747304  random=0.748853
n=16:  left=0.766338  right=0.726831  mid=0.746944  random=0.751727
n=32:  left=0.756641  right=0.736887  mid=0.746854  random=0.747644
n=64:  left=0.751748  right=0.741871  mid=0.746832  random=0.747008
n=128:  left=0.749290  right=0.744351  mid=0.746826  random=0.746846
n=256:  left=0.748058  right=0.745589  mid=0.746825  random=0.746818
n=512:  left=0.747441  right=0.746207  mid=0.746824  random=0.746821
n=1024:  left=0.747133  right=0.746515  mid=0.746824  random=0.746814
n=2048:  left=0.746978  right=0.746670  mid=0.746824  random=0.746821
n=4096:  left=0.746901  right=0.746747  mid=0.746824  random=0.746824
n=8192:  left=0.746863  right=0.746786  mid=0.746824  random=0.746824
n=16384:  left=0.746843  right=0.746805  mid=0.746824  random=0.746824
n=32768:  left=0.746834  right=0.746814  mid=0.746824  random=0.746824
n=65536:  left=0.746829  right=0.746819  mid=0.746824  random=0.746824
n=131072:  left=0.746827  right=0.746822  mid=0.746824  random=0.746824
n=262144:  left=0.746825  right=0.746823  mid=0.746824  random=0.746824
n=524288:  left=0.746825  right=0.746824  mid=0.746824  random=0.746824
```

Figure 1: Question 1(c)

$$f(0)=1, \quad f'(0)=\tfrac{1}{2}, \quad f''(0)=-\tfrac{1}{4}, \quad f^{(3)}(0)=\tfrac{3}{8}.$$

$$P_3(x)=f(0)+f'(0)\,x+\frac{f''(0)}{2!}x^2+\frac{f^{(3)}(0)}{3!}x^3=1+\frac{1}{2}x-\frac{1}{8}x^2+\frac{1}{16}x^3.$$

y	$x=y-1$	$P_3(x)$	$\sqrt{y}$	$\lvert P_3(x)-\sqrt{y}\rvert$
0.50	−0.50	0.7109375000	0.7071067812	0.0038307188
0.75	−0.25	0.8662109375	0.8660254038	0.0001855337
1.25	0.25	1.1181640625	1.1180339887	0.0001300738
1.50	0.50	1.2265625000	1.2247448714	0.0018176286

Table 1: Question 2

3. **Convergence of the Bisection Method**

   Let $f \in C([a,b])$ with $f(a)>0$, $f(b)<0$. Let (p_n) be the sequence generated by the Bisection Method to approximate a root p of f.

   (a) Show that after one step, $|p-p_1| \le (b-a)/2$. The first bisection iterate is the midpoint $p_1=\frac{a+b}{2}$. For any $x\in[a,b]$,

   $$\left|x-\frac{a+b}{2}\right| \le \max\left\{\left|a-\frac{a+b}{2}\right|,\ \left|b-\frac{a+b}{2}\right|\right\}=\frac{b-a}{2}.$$

   Since $p\in[a,b]$, substituting $x=p$ yields

   $$|p-p_1| \le \frac{b-a}{2}.$$

   (b) By repeating the argument, show that after n steps, $|p-p_n| \le (b-a)/2^n$. At each step the interval length halves so that

   $$b_{n-1}-a_{n-1}=\frac{b-a}{2^{\,n-1}}.$$

   Since $p, p_n \in [a_{n-1}, b_{n-1}]$ and p_n is the midpoint, the distance from p_n to any point in this interval is at most half its length. Hence

   $$|p-p_n| \ \le\ \frac{b_{n-1}-a_{n-1}}{2}\ =\ \frac{1}{2}\cdot\frac{b-a}{2^{\,n-1}}\ =\ \frac{b-a}{2^{\,n}}.$$

   Therefore, after n steps,

   $$|p-p_n| \le \frac{b-a}{2^{\,n}}.$$

(c) Conclude that $p_n \to p$ as $n \to \infty$. Since $\frac{b-a}{2^n} \to 0$ as $n \to \infty$, it follows that $|p - p_n| \to 0$. Hence $p_n \to p$ as $n \to \infty$.